

Engineering Research Initiative: A Proof-of-Concept LoRa Mesh Network for Inter-Campus Communication Redundancy

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Abstract

This study proposes the development of a Phase 1 experimental Long-Range (LoRa) mesh radio network to serve as a high-reliability research testbed for the University of North Georgia. Utilizing the Meshtastic open-source protocol on the 915 MHz ISM band, the project seeks to establish a decentralized data link between the Gainesville (GVL) and Dahlonega (DAH) campuses. This initial phase focuses exclusively on validating RF propagation models and hardware durability in North Georgia's terrain without interacting with official university data systems. The results of this proof-of-concept will provide the empirical data necessary for the UNG Department of Public Safety to evaluate the technology for future emergency communication redundancy in a subsequent Phase 2 assessment.

1 Introduction

In November 18, 2025, global outages disrupted the internet through a Cloudflare failure, one of the largest internet service providers [4]. January 14th, 2026, over a million people were affected by a major Verizon network outage [5]. Early in 2026, two major winter storms caused power failures throughout Georgia and statewide shutdowns [3]. These events lead to members of the Engineering Student Guild in UNG Gainesville to explore alternative, decentralized, and reliable

communication solutions. Meshtastic is an open-source wireless communication system that utilizes long range (LoRa) radio frequencies 915 MHz in the US[2]. It is built on a node system, a mesh network of radios relaying messages without a central gateway. Meshtastic would benefit UNG, providing a reliable communication infrastructure connecting all campuses, immune to failure modes from centralized networks such as ransomware, catastrophic events, or internet dependency. This proposal details an initial proof of concept to evaluate a Meshtastic implementation as a redundancy layer to UNG Department of Safety's communication system, through field testing and network analysis of a prototype link between Dahlonega and Gainesville campuses. A minimum viable design will provide a framework for future implementations across all campuses.

2 Technical Framework and RF Propogation Analysis

2.1 Architecture

Designing an architecture for inter-campus communication must take the aforementioned radio parameters and the topology of North Georgia.

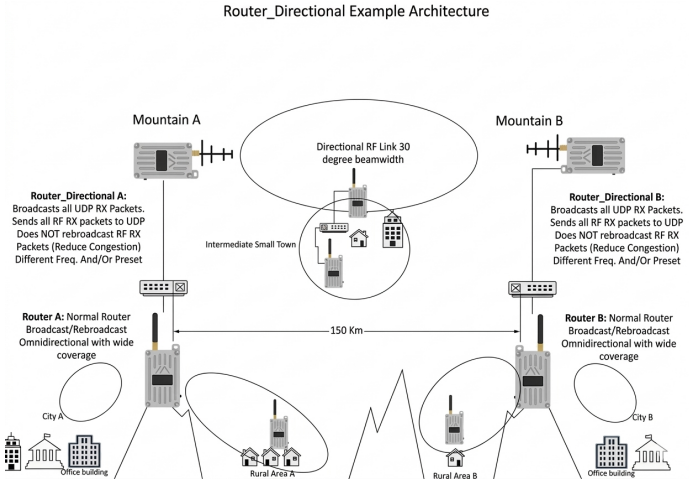


Figure 1: Architecture example

Since height is the determining factor in wave propagation, a directional router design leverages North Georgia's mountainous terrain to increase signal distance. As a proof of concept, a mesh node on Yonah Mountain, Cleveland is utilized as a central hub node between Dahlonega and Gainesville. Each respective campus operates a omni-directional antenna for local broadcasting, and Yagi directional antennas aimed at hub nodes for connecting campuses.

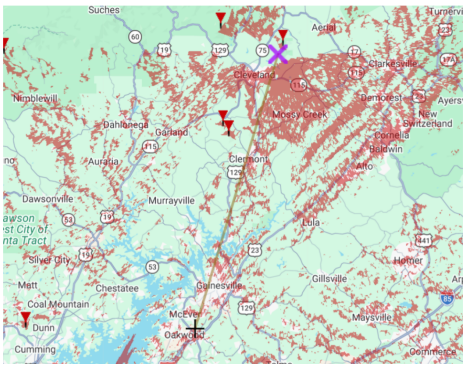


Figure 2: GVL signal availability in red

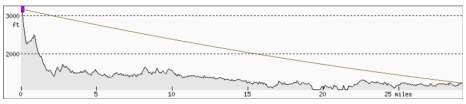


Figure 3: Yonah to GVL LOS

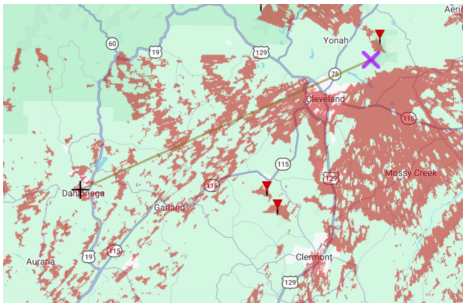


Figure 4: DAH signal availability in red

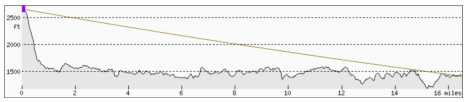


Figure 5: Yonah to DAH LOS

2.1.1 Dual-Path Antenna Configuration

To achieve a link, each node operates two distinct signal paths:

- **Backhaul Link (Directional):** A concentrated, high-gain beam is aimed at a central relay such as the Yonah Mountain node. 12 dBI Yagi antennas minimize signal disparity, overcoming atmospheric attenuation and foliage interference. This signal path relays campuses over large distances.
- **Local Broadcast (Omni-Directional):** Dipole antennas provide immediate campus coverage, allowing research nodes and handhelds to interface with system from any direction.

2.2 Propagation Engineering

The design addresses physical constraints specific to North Georgia's topology.

- **Radio Horizon:** Height is a determining factor in wave propagation. Roof mounts extend the geometric horizon to ≈ 22.5 km, clearing topographical obstructions between GVL/DAH and the hub node.
- **Link Budget and EIRP Compliance:** The system is tuned to a maximum transit power of 22dBm [6]. The system should operate at an EIRP of 33 dBm, compliant with FCC Part 15.247 [1].

3 Evaluation Criteria

3.1 Link Quality and Reliability Metrics

The standard for radio links performance is its ability to deliver data without corruption.

- **Packet Delivery Ratio (PDR):** The ratio of packets successfully received to total packets transmitted.

$$\text{PDR} = \frac{\sum \text{Packets Received}}{\sum \text{Packets Transmitted}} \times 100 \tag{1}$$

  A success rate of $\geq 98\%$ is ideal.
- **Received Signal Strength Indicator (RSSI):** Measured in dBm. This indicates the power level being received by the antenna.
- **Signal-to-Noise Ratio (SNR):** Critical for LoRa, as it can decode signals below the noise floor. A consistent SNR above -10dB is required for stable "Long-Fast" settings.

3.2 Environmental and Power Autonomy

- **Battery Discharge Cycle:** Gauge State of Charge (SoC) and battery voltage over 72-hour periods of no solar gain.
- **Thermal Stability:** Monitoring internal enclosure temperatures against ambient North Georgia temperatures to ensure the LuckFox Ultra processing units remain within the operating range of $-20°C$ to $70°C$.

4 Project Scope and Phasing

To ensure technical rigor and institutional security, this project is divided into two distinct phases. This proposal seeks funding and approval for **Phase 1** only.

4.1 Phase 1: Technical Proof of Concept (SGA Funded)

This phase focuses on the establishment of a student-led node network for research purposes.

- **Hardware Deployment:** Installation of research nodes at GVL and DAH campuses.
- **Data Isolation:** No UNG sensitive data or Public Safety systems will interact with this network. Students will use independent devices to test latency and PDR.
- **Performance Analysis:** Evaluation of the link budget and signal reliability over the North Georgia terrain.

4.2 Phase 2: Public Safety Evaluation (Future Scope)

Upon successful completion of Phase 1, the infrastructure will be made available to the UNG Department of Public Safety for a formal evaluation. This will include:

- Full software security assessment and encryption key management by Public Safety staff.
- Integration into official continuity plans and maintenance schedules.

5 Phase 1

5.1 Design Constraints

- **System Autonomy:** In Phase 1, university systems and data do not interact with the prototype. Sensitive information cannot be transmitted or received through node channels.
- **Power Autonomy:** To serve the function of a truly redundant layer, the node's power will be sourced independently of university grid and power systems.
- **Offline:** The system will be self-contained, never interacting with or connecting to online gateways and servers, neither public nor private. All data will be stored locally, and the only wireless signals permissible are 915MHz LoRa or 2.4GHz bluetooth protocols for internode communication or local bluetooth access, respectively.

- **Encryption:** No encryption, encryption keys, or passwords will be used or are necessary for the Phase 1 design.
- **Non-invasive Installation:** No boring, drilling, or damage to university property will be required for installation of the nodes. Free-standing antenna masts or clamps are ideal.
- **Mesh Utilization:** For the initial proof-of-concept, the local mesh and other nodes, such as the Yonah Mountain relay, will be used temporarily for testing in Phase 1. The mesh helps establish a baseline for evaluation without increasing design costs. In Phase 2, a proprietary mesh could be designed with Phase 1 data.

5.2 Deliverables

- **Report:** A technical test report with a detailed analysis of every performance metric and functionality.
- **Configuration Documentation:** Full configuration documentation, firmware versions, LoRa Presets, and power system specifications per node.
- **Site Documentation:** including installation and mechanic drafts, photographs, and standard of operations.

6 Phase 1 Budget and Bill of Materials (Phase 1)

Table 1: Bill of Materials - Per Node Location

Component	QTY	Unit Price	Total	Notes
5.8 dBi N-Male Omni Antenna	2	$42.97	$85.94	Local coverage
ALFA 12 dBi Directional Yagi	1	$64.97	$64.97	Backhaul link
LuckFox Ultra Hat V3 1W	1	$43.00	$43.00	High-power radio
LuckFox Lyra Ultra (PoE/WiFi)	1	$44.99	$44.99	Processing unit
NULLHOP MeshToad V3	1	$45.00	$45.00	Carrier board
UFL to N-Female Pigtail	1	$6.97	$6.97	Bulkhead mount
UFL to N-Male Pigtail	1	$6.97	$6.97	Internal cabling
Ubiquiti TOUGHCable PRO (Shielded)	1	$165.60	$165.60	Outdoor rated
M20 RJ45 Waterproof Connectors	1	$16.99	$16.99	IP67 ingress prot.
Redodo 12V 50Ah LiFePO4 Battery	1	$118.99	$118.99	Power autonomy
Renogy 50W Solar Panel + Mount	1	$120.98	$120.98	Off-grid supply
LINOVISION Industrial PoE Switch	1	$85.99	$85.99	12V to 48V boost

Table 1 continued from previous page

Component	QTY	Unit Price	Total	Notes
Victron SmartSolar MPPT (75/15)	1	$64.99	$64.99	Bluetooth mgmt
Radio Enclosure (Internal)	1	$14.50	$14.50	Secondary housing
Stainless Waterproof Vent Valves	1	$8.83	$8.83	Pressure equalization
Universal Enclosure Mounting Kit	1	$86.99	$86.99	Tower/Mast mount
Altelix 14x12x8 NEMA Enclosure	1	$224.99	$224.99	Fiberglass/Vented
Nebra Repeater Arm Mount	1	$70.00	$70.00	Custom bracket
Non-Penetrating Roof Mount (10')	1	$673.62	$673.62	**No-drill install**
HSS 5/8"-24 Tap and Die Set	2	$15.99	$31.98	Custom mounting
Misc (Shipping, Supplies, Tax)	1	$600.00	$600.00	Contingency
Total Estimated Cost			**$2,582.29**	**Per Node**

References

[1] *FCC Code Regulations*. URL: https://www.ecfr.gov/current/title-47/chapter-I/subchapter-A/part-15/subpart-C/subject-group-ECFR2f2e5828339709e/section-15.247.

[2] *Meshtastic*. URL: https://meshtastic.org/docs.

[3] *NOAA Satellites Monitor Massive Winter Storm*. Tech. rep. NOAA, 2026. URL: https://www.nesdis.noaa.gov/news/noaa-satellites-monitor-massive-winter-storm.

[4] Matthew Prince. *Cloudflare outage on November 18, 2025*. 2025. URL: https://blog.cloudflare.com/18-november-2025-outage/.

[5] John Ruwitch. *Verizon just had a big outage. Here's what we know*. 2026. URL: https://www.npr.org/2026/01/15/nx-s1-5678889/verizon-outage-what-happened.

[6] *What is a Link Budget?* URL: https://www.mathworks.com/discovery/link-budget.html.